ASSISTANT PROFESSOR EXAM — 2024

Economics • Second Paper

UNIT 9

Statistical Methods

सांख्यकीय वधियाँ

1. Central Tendency & Dispersion, Skewness & Kurtosis; Correlation & Regression; Index Numbers & Time Series

2. Sampling & Census; Binomial, Poisson & Normal Distribution; Random Variable & Probability; Hypothesis Testing; t / F / χ^2 / ANOVA

3. CLRM — Assumptions & Properties; Multiple Regression Model: Estimation & Interpretation

4. Application of Differential & Integral Calculus and Matrix in Consumer Behaviour, Production & Pricing

5. Input-Output Analysis & Linear Programming

Hinglish detailed notes • 16 diagrams • formulae + worked numericals • exam-oriented

SECTION 1 — Descriptive Statistics, Correlation, Regression, Index & Time Series

1.1 Measures of Central Tendency / केन्द्रीय प्रवृत्ति

Concept: Aisa single value jo entire distribution ko represent kare — "typical" observation. Statistician Yule ke per ideal average ke 6 properties: rigid definition, easy to compute, based on all observations, capable of further algebraic treatment, not unduly affected by extreme values, sampling stability.

Measure	Formula (ungrouped)	Property / Use
Arithmetic Mean ($\bar{X}$)	$\bar{X} = \Sigma X / N$	most common; affected by extremes; sum of deviations = 0
Weighted Mean	$\bar{X}_w = \Sigma(wX)/\Sigma w$	when items unequal importance — index numbers
Median (M)	size of (N+1)/2-th item	positional; not affected by extremes; for open-end classes
Mode (Z)	most frequent value	fashion item; ready-made garments, footwear sizes
Geometric Mean	$GM = (X_1 \cdot X_2 \ldots X_n)^{1/n}$	rates of growth, index numbers — relative change
Harmonic Mean	$HM = N / \Sigma(1/X)$	rates and ratios (speed) — averages of rates

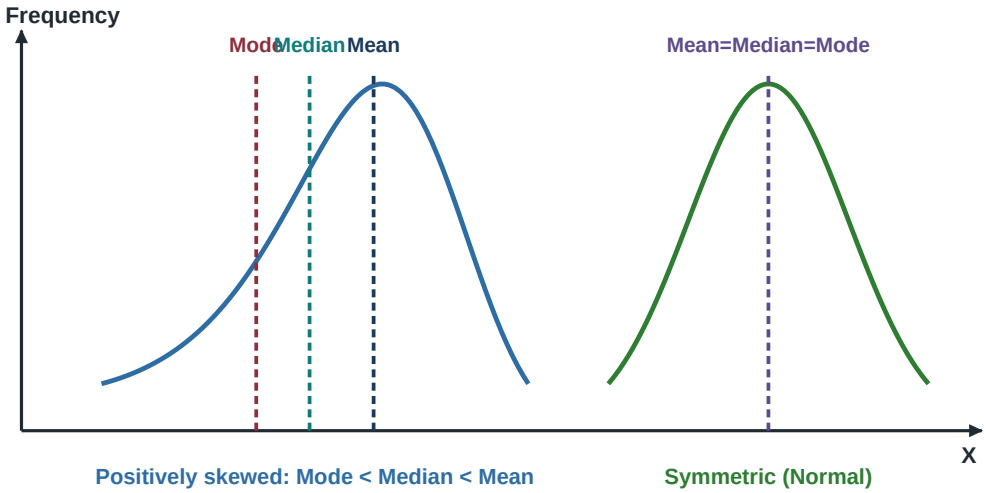


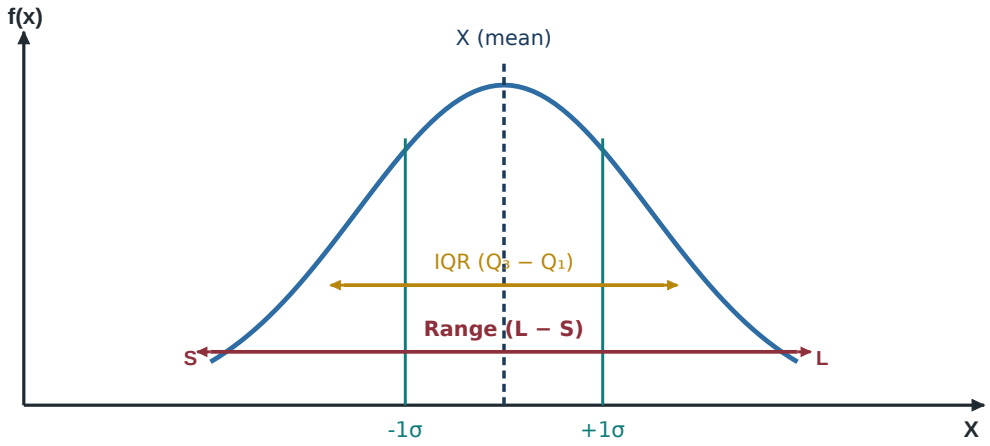
Fig 1: Mean–Median–Mode positions on skewed vs symmetric distribution

Empirical relation (moderately skewed)
Mode = 3 × Median − 2 × Mean (Karl Pearson). AM ≥ GM ≥ HM (with equality only when all values are equal).

Worked example — Central tendency
Data: 5, 7, 8, 10, 15, 18, 25. n = 7. Mean = 88/7 = 12.57; Median = (n+1)/2 = 4th item = 10; Mode = none (no repetition). GM = $(5 \cdot 7 \cdot 8 \cdot 10 \cdot 15 \cdot 18 \cdot 25)^{1/7}$ ≈ 11.39. HM = 7 / (1/5+1/7+1/8+1/10+1/15+1/18+1/25) ≈ 9.97. Note: AM > GM > HM ✓

1.2 Measures of Dispersion / प्रसरण / वषिमता

Concept: Spread/scatter of values around central value. Average alone insufficient — do groups same mean ke ho sakte hain par very different spread. Properties (Yule): rigidly defined, easy to calculate, based on all observations, algebraic treatment, not unduly affected by extremes.



Absolute: Range, IQR, MD, SD | Relative: CV, QD coefficient, MD coefficient

Fig 2: Measures of dispersion on a frequency distribution

Measure	Formula	Comment
Range (R)	R = L − S (Largest − Smallest)	simplest; affected only by extremes; used in QC charts
Quartile Deviation (QD)	QD = (Q₃ − Q₁) / 2	semi-IQR; for open-end / skewed data; covers middle 50%
Mean Deviation (MD)	MD = Σ|X − Mean| / N	absolute deviations; usually about median
Variance (σ^2)	$\sigma^2 = \Sigma(X - \bar{X})^2 / N$	in squared units; basis for SD
Standard Deviation (σ, s)	$\sigma = \sqrt{[\Sigma(X - \bar{X})^2 / N]}$	"king of dispersion measures"; same units as data
Coefficient of Variation (CV)	CV = (σ / $\bar{X}$) × 100	relative measure — compare groups with different units

Numerical — SD and CV

Data: 4, 6, 8, 10, 12. $\bar{X}$ = 8. Deviations: −4, −2, 0, 2, 4. Squared: 16, 4, 0, 4, 16 → Σ = 40. σ^2 = 40/5 = 8; σ = √8 = 2.83. CV = (2.83/8)×100 = **35.4%**. Use CV jab do datasets ki units alag ho (e.g., agriculture yield vs height of plants).

1.3 Skewness / वक्रिमता

Skewness = lack of symmetry. **Direction**: tail kis taraf lambi hai — wahi skew ki direction. **Symmetry** me Mean = Median = Mode.

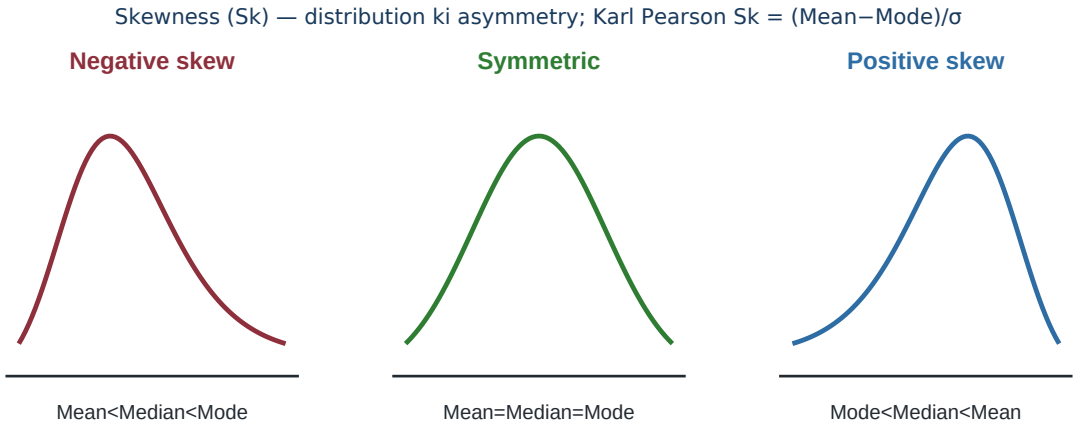


Fig 3: Three skewness types

Skewness measures

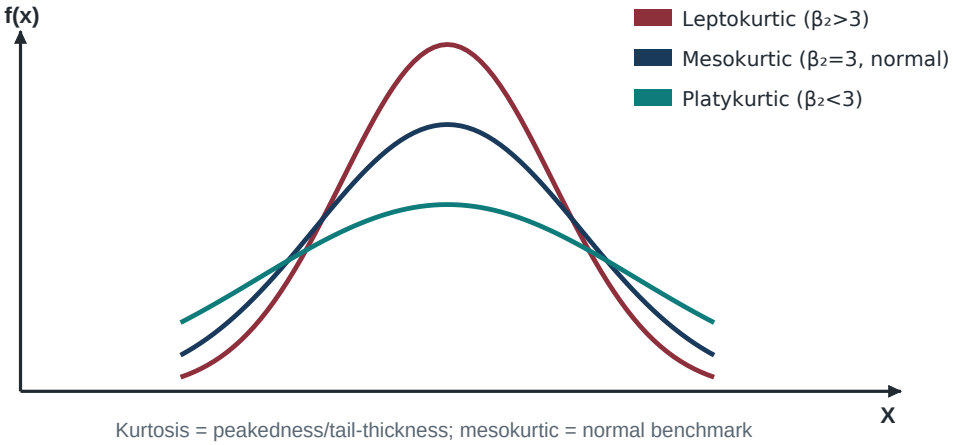
Karl Pearson Sk = (Mean − Mode) / σ; if mode ill-defined: Sk = 3(Mean − Median)/σ. Range: typically −3 to +3.

Bowley's Sk (Quartile-based) = [(Q₃ + Q₁ − 2·Median)] / (Q₃ − Q₁). Range −1 to +1.

Moment-based $\beta_1 = \mu_3^2 / \mu_2^3$; $\gamma_1 = \sqrt{\beta_1}$ (sign of μ_3 gives direction).

1.4 Kurtosis / कुकुदता

Kurtosis = peakedness aur tail-thickness ka measure. Compared to normal (mesokurtic).



Kurtosis = peakedness/tail-thickness; mesokurtic = normal benchmark

Fig 4: Three kurtosis types (lepto / meso / platy)

Kurtosis metric

$\beta_2 = \mu_4 / \mu_2^2$; Excess kurtosis $\gamma_2 = \beta_2 - 3$. $\beta_2 > 3$ → leptokurtic (peaked, fat tails — e.g., financial returns); $\beta_2 = 3$ → mesokurtic (normal benchmark); $\beta_2 < 3$ → platykurtic (flat).

SECTION 1 (cont.) — Correlation, Regression, Index & Time Series

1.5 Correlation / सहसम्बन्ध

Correlation = degree and direction of *linear* association between two variables X and Y. **Yule**: "Correlation analysis enables us to have an idea about the degree and direction of relationship." Note: **correlation ≠ causation** (spurious correlations, common causes).

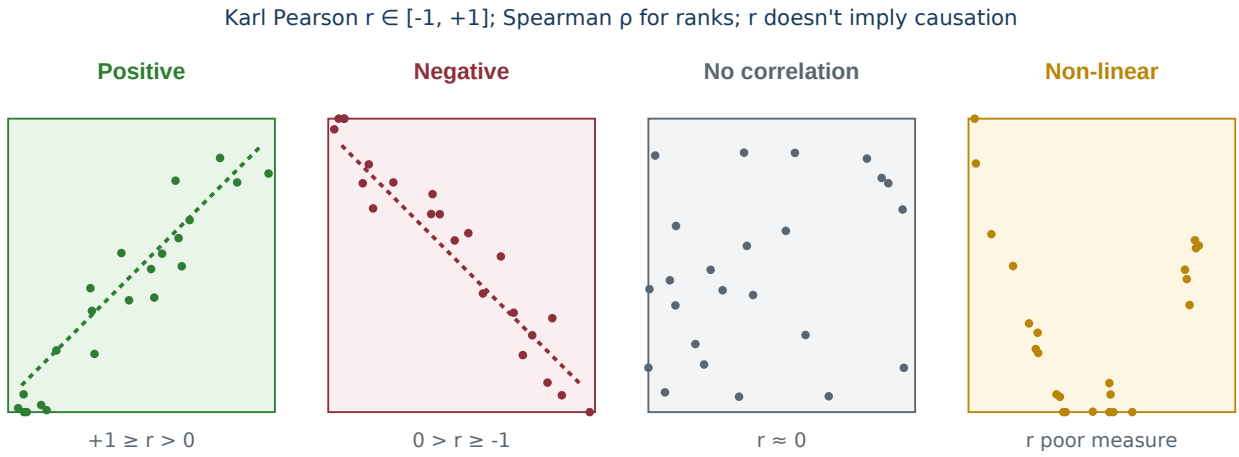


Fig 5: Four scatter patterns

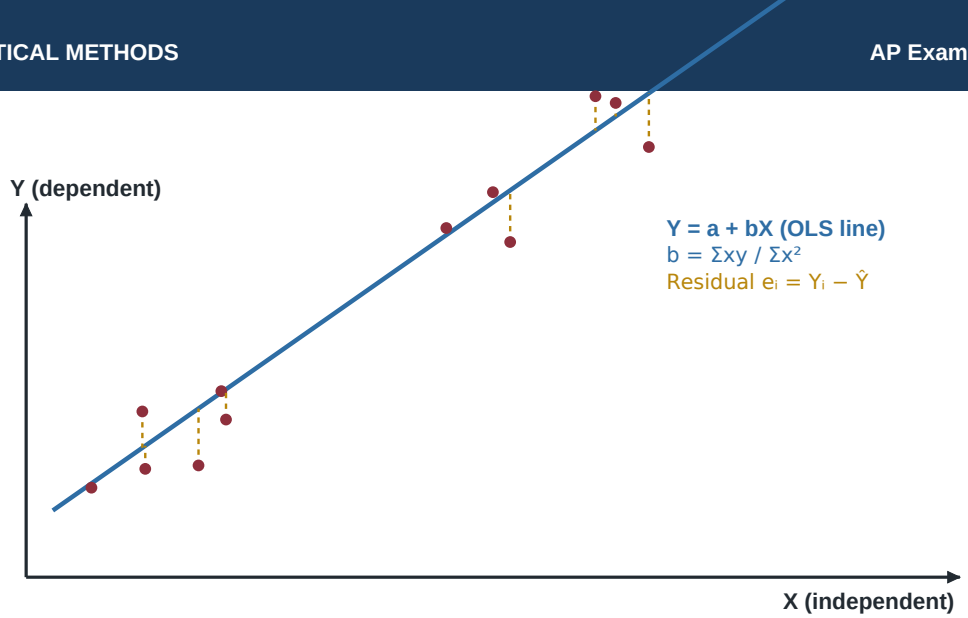
Method	Formula	When to use
Karl Pearson r	$r = \Sigma xy / \sqrt{(\Sigma x^2 \cdot \Sigma y^2)}$ where $x = X - \bar{X}$, $y = Y - \bar{Y}$	quantitative + linear; assumes interval data, normal-ish
Spearman ρ (rank)	$\rho = 1 - 6\Sigma d^2 / [n(n^2-1)]$	qualitative / ordinal data; outliers ka effect kam
Concurrent Deviation	$r_c = \pm\sqrt{[(2c - n) / n]}$	quick check — c = no. of concurrent (+) deviations
Coefficient of Determination	r^2 (range 0 to 1)	% variation in Y explained by X

Properties of r (Karl Pearson)

- $-1 \le r \le +1$; r = +1 perfect positive, r = −1 perfect negative, r = 0 no linear relation.
- r is a **pure number** — independent of unit and origin (i.e., shift / scale change ka asar nahi).
- Geometric mean of two regression coefficients: $r = \sqrt{(b_{yx} \cdot b_{xy})}$.
- Probable Error: $PE = 0.6745 \times (1 - r^2) / \sqrt{n}$; $r > 6 \cdot PE \rightarrow$ significant.

1.6 Regression Analysis / प्रतगिमन

Galton (1886) ki "regression to the mean" se concept aaya. **Difference vs correlation:** correlation = symmetric (X aur Y interchangeable); regression = **asymmetric** (X explains Y). Regression batata hai **kitna** change Y me hota hai for unit change in X.



OLS minimises Σe_i^2; $b_{yx} \times b_{xy} = r^2$; both lines intersect at (X, Y)

Fig 6: OLS regression line with residuals

Two regression lines

Y on X: $Y - \bar{Y} = b_{yx}(X - \bar{X})$, $b_{yx} = \Sigma xy / \Sigma x^2 = r \cdot (\sigma_y/\sigma_x)$

X on Y: $X - \bar{X} = b_{xy}(Y - \bar{Y})$, $b_{xy} = \Sigma xy / \Sigma y^2 = r \cdot (\sigma_x/\sigma_y)$

• Both lines intersect at $(\bar{X}, \bar{Y})$. • $r^2 = b_{yx} \cdot b_{xy}$. • Same sign as r.

• OLS minimises Σe_i^2 where $e_i = Y_i - \hat{Y}_i$ (residual).

Worked numerical — Find r from regression coefficients

Given $b_{yx} = 0.8$ and $b_{xy} = 0.45$. Then $r = \sqrt{(0.8 \times 0.45)} = \sqrt{0.36} = \mathbf{0.6}$ (positive — both b's positive). Interpretation: 36% of Y's variation is explained by X ($R^2 = 0.36$).

1.7 Index Numbers / सूचकांक

Index number = specialised average that measures relative change in a variable / group of variables over time, geography, or category. **Fisher**: "Index numbers are the economic barometers." Examples: CPI, WPI, IIP, BSE Sensex, HDI.

Formula	Expression	Note
Simple Aggregative	$P_{01} = (\Sigma p_1 / \Sigma p_0) \times 100$	price relative; ignores quantity weights
Laspeyres (L)	$L = (\Sigma p_1 q_0 / \Sigma p_0 q_0) \times 100$	**base-year quantities**; upward bias
Paasche (P)	$P = (\Sigma p_1 q_1 / \Sigma p_0 q_1) \times 100$	**current-year quantities**; downward bias
Fisher's Ideal (F)	$F = \sqrt{(L \times P)}$	GM of L & P; satisfies time and factor reversal tests
Marshall-Edgeworth	$(\Sigma p_1(q_0+q_1) / \Sigma p_0(q_0+q_1)) \times 100$	average quantities as weights
Walsh	$(\Sigma p_1\sqrt{(q_0 q_1)} / \Sigma p_0\sqrt{(q_0 q_1)}) \times 100$	GM of quantities as weights

Tests of an ideal index

(1) Unit test — index independent of units (all except simple aggregative pass).

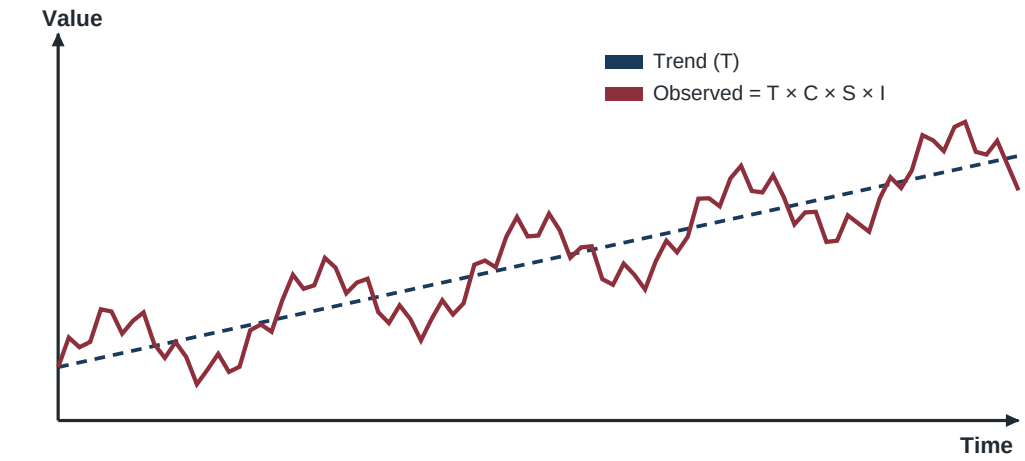
(2) Time Reversal: $P_{01} \times P_{10} = 1$ (Fisher passes).

(3) Factor Reversal: $P_{01} \times Q_{01} = V_{01}$ (value ratio). Only Fisher passes.

(4) Circular test: $P_{01} \cdot P_{12} \cdot P_{20} = 1$ (simple GM & Marshall pass; Fisher does NOT).

1.8 Time Series Analysis / काल श्रेणी

Time series = data observed at successive equal time intervals. Components — TCSI.



Components: T (Trend), C (Cyclical), S (Seasonal), I (Irregular). Models: Additive Y=T+C+S+I; Multiplicative Y=T×C×S×I

Fig 7: Time series — observed = T × C × S × I (multiplicative model)

Four components

Trend (T) — long-term smooth movement (e.g., GDP rising over decades). Methods: free-hand, semi-averages, moving averages, least squares.

Cyclical (C) — oscillations > 1 year (business cycle: ~5–10 yrs). Phases: prosperity, recession, depression, recovery.

Seasonal (S) — within a year (festive demand, monsoon agriculture). Measured by simple averages, link relatives, ratio-to-trend / ratio-to-MA.

Irregular (I) — random / erratic (war, pandemic, drought). Unpredictable, isolated by elimination.

Models: Additive: Y = T + C + S + I (independent components); Multiplicative: Y = T × C × S × I (proportional).

KEY TERMS / परिभाषाएँ

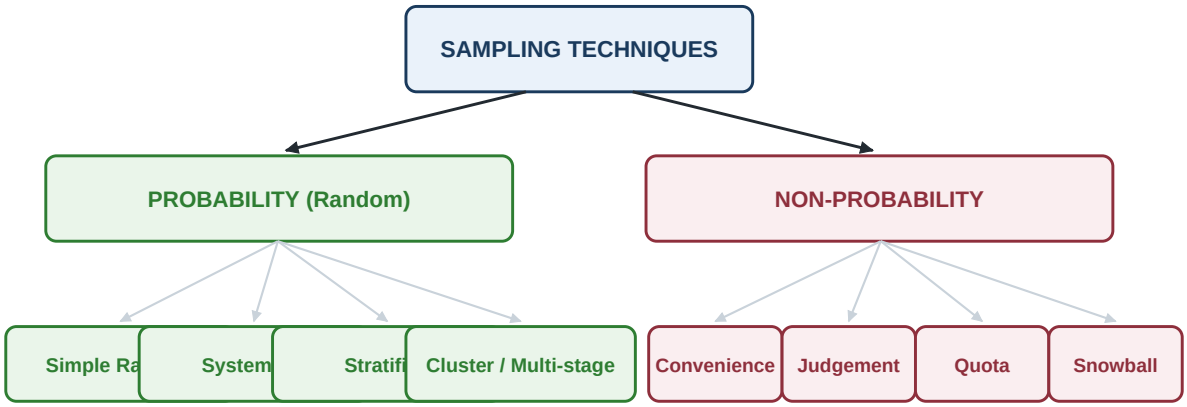
CPI	Consumer Price Index — retail inflation; base 2012 (current series); CPI-IW, CPI-AL, CPI-RL, CPI-C/R/U
WPI	Wholesale Price Index — base 2011-12; used by RBI till 2014, now CPI for monetary policy
IIP	Index of Industrial Production — base 2011-12; sectors: mining, manufacturing, electricity
HDI	Human Development Index — composite of income, education, life expectancy (UNDP)
Splicing	connecting old & new base series of an index for continuity
Chain Index	$P_{0n} = P_{01} \times P_{12} \times \ldots \times P_\{n-1,n\}$ — circular test pass; flexible weights

SECTION 2 — Sampling, Probability Distributions, Hypothesis Testing

2.1 Sampling vs Census / नदिर्श्न एवं संगणना

Census = complete enumeration (every unit surveyed; n = N). India ka decadal Population Census example. **Sampling** = study of a representative subset to infer about population. Both have merits & limits.

Aspect	Census	Sampling
Coverage	Entire population	Subset (representative)
Accuracy	100% (if no error)	Sampling error present but estimable
Cost & Time	Very high	Lower; faster
Use case	small population, regulatory needs	large pop, destructive testing, periodic surveys



Census = complete enumeration (n=N); Sampling = subset survey — saves cost, time

Sampling errors (estimable) vs Non-sampling errors (response, measurement bias)

Fig 8: Classification of sampling techniques

Type	Description	Example
Simple Random	every unit equal probability; lottery / random number tables	select 50 students from 500
Systematic	every k-th unit from a list, random start (k = N/n)	every 10th household in a colony
Stratified	population divided into homogeneous strata, then random within	urban/rural, gender, income groups
Cluster / Multi-stage	natural clusters (villages, schools); sample clusters	NSSO village sampling
Convenience	easy / handy units — biased	street interviews
Quota	fixed quota per category, then non-random pick	opinion polls
Snowball	one respondent refers next — hidden populations	drug users, migrants

Errors in sample surveys

Sampling Error — difference between sample stat and population parameter due to chance; decreases as n increases (SE = σ/√n).

Non-sampling Errors — measurement, response (lies), non-response, data-entry, framing bias. Census me bhi present; sample me bhi.

2.2 Random Variable & Probability Distribution / यादृच्छकि चर

Random variable (X) = a real-valued function defined on the sample space of a random experiment. **Discrete** (countable: 0,1,2,…) vs **Continuous** (uncountable: any real in an interval).

Key concepts

PMF (discrete): $P(X = x) \geq 0$; $\Sigma P(x) = 1$. **PDF** (continuous): $f(x) \geq 0$; $\int f(x)dx = 1$ over support.

CDF: $F(x) = P(X \leq x)$. **Expectation**: $E(X) = \Sigma x \cdot P(x)$ (discrete) or $\int x \cdot f(x)dx$ (continuous).

Variance: $Var(X) = E(X^2) - [E(X)]^2$. **Moments**: r-th central moment $\mu_r = E[(X - \mu)^r]$.

2.3 Three Important Distributions

Distribution	PMF / PDF	Mean & Variance / Notes
Binomial B(n,p)	$P(X=r) = {}^nC_r\, p^r(1-p)^{n-r}$	Mean = np, Var = npq. Discrete; n trials; success prob p. Example: 10 tosses, P(7 heads).
Poisson P(λ)	$P(X=r) = e^{-\lambda}\lambda^r/r!$	Mean = Var = λ. Discrete; rare events in fixed interval. Limit of Binomial when $n\to\infty$, $p\to0$, $np = \lambda$. Example: customers per minute at counter.
Normal N(μ, σ²)	$f(x) = (1/\sigma\sqrt{(2\pi)})\, e^{-(x-\mu)^2/(2\sigma^2)}$	Mean = Median = Mode = μ. Continuous; bell-shaped, symmetric. $Z = (X-\mu)/\sigma$ standardises any normal. Most important — CLT basis.

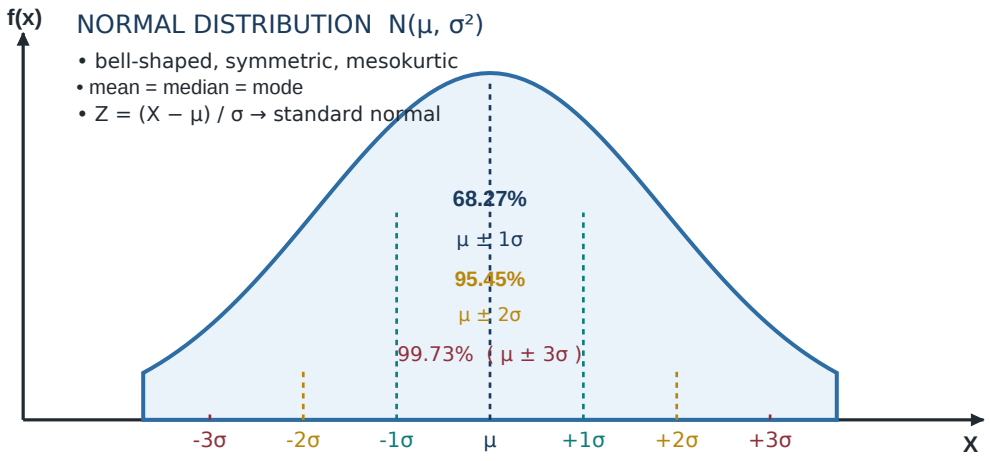


Fig 9: Normal distribution — empirical rule (68-95-99.7)

Worked numerical — Normal distribution

IQ scores follow N(μ=100, σ=15). P(IQ > 130)? $Z = (130-100)/15 = 2$. $P(Z > 2) = 0.5 - 0.4772 =$ **0.0228** (i.e., ~2.28% population). 95% of IQ ki range: $\mu \pm 1.96\sigma = 100 \pm 29.4 \rightarrow [70.6, 129.4]$.

Central Limit Theorem (CLT)

Statement: Any population (mean μ, finite variance σ²) se random samples of size n lo. For $n \geq 30$, the sampling distribution of $\bar{X}$ is approximately Normal with mean μ and variance σ^2/n — **irrespective of the shape of the parent distribution**. CLT is the foundation of statistical inference.

SECTION 2 (cont.) — Hypothesis Testing, Errors, t / F / χ^2 / ANOVA

2.4 Hypothesis Testing / परिकल्पना परीक्षण — Logic

Statistical hypothesis = an assumption about a population parameter, which we test using sample evidence. Two types: **Null hypothesis H_0** (no effect / no difference — to be tested) vs **Alternative H_1** (research hypothesis — what we suspect is true).

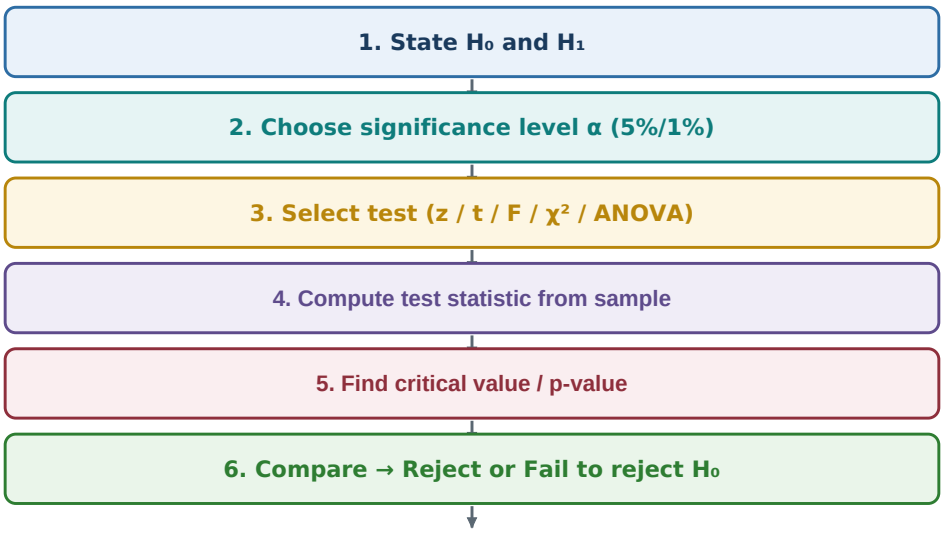


Fig 10: Six steps of hypothesis testing

	H_0 is TRUE	H_0 is FALSE
Reject H_0	**TYPE I ERROR (α)** Reject true H_0	**CORRECT** (Power = 1−β)
Accept H_0	**CORRECT** (Confidence = 1−α)	**TYPE II ERROR (β)** Accept false H_0

α = significance level; β = type-II prob.; Power = 1−β — sample size badhane se dono kam

Trade-off: α ghata-do to β badh jaata hai (with fixed n)

Fig 11: Type I vs Type II error — confusion matrix

Errors in hypothesis testing

Type I Error (α) = Reject H_0 when it is actually **true** (false positive). α = level of significance, usually 0.05 or 0.01.

Type II Error (β) = Accept H_0 when it is actually **false** (false negative).

Power of test = 1 − β = probability of correctly rejecting a false H_0. Sample size n badhane se dono α, β kam ho sakte hain (trade-off relaxes).

p-value = probability of observing the test statistic (or more extreme) **given H_0 is true**. Decision rule: reject H_0 if p-value < α.

2.5 Test Statistics — t, F, χ^2, ANOVA

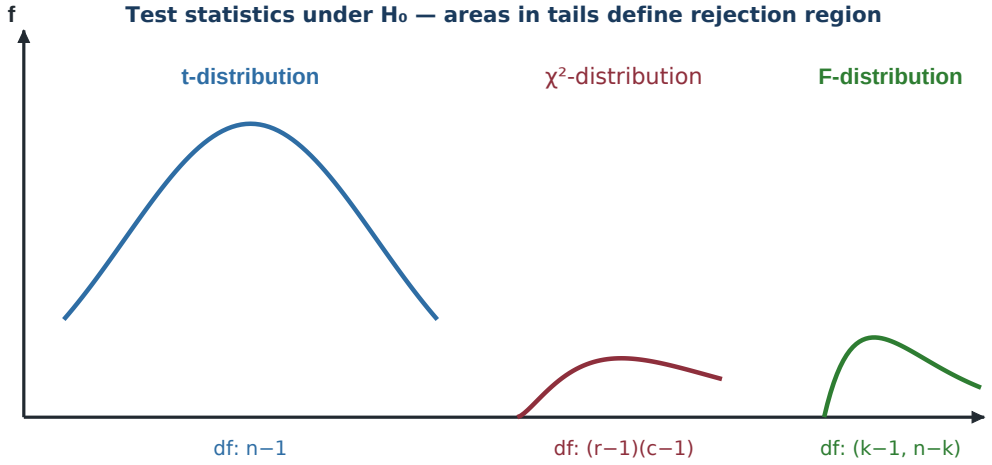


Fig 12: Shapes of t / χ² / F distributions

Test	Use	Formula / df
Z-test	large sample (n ≥ 30), σ known; test of mean / proportion	$Z = (\bar{X} - \mu_0) / (\sigma/\sqrt{n})$
t-test (Gosset 1908)	small sample (n < 30), σ unknown; one-sample / two-sample / paired	$t = (\bar{X} - \mu) / (s/\sqrt{n})$; df = n − 1
Chi-square (χ^2) test	(a) goodness of fit; (b) independence of attributes (contingency table); (c) test of variance	$\chi^2 = \Sigma[(O - E)^2 / E]$; df = (r−1)(c−1)
F-test	compare two variances; significance of regression	$F = s_1^2 / s_2^2$ (larger / smaller); df: (n_1−1, n_2−1)
ANOVA (Fisher)	compare means of 3+ groups simultaneously (extends t-test)	$F = MS_{Between} / MS_{Within}$; df: (k−1, n−k)

ANOVA — total variance partition

SST = SSB + SSW (Total = Between groups + Within groups).

• **MSB** = SSB / (k − 1); **MSW** = SSW / (n − k). • **F = MSB / MSW**.

• If $F > F_{critical}$ → reject H_0 (at least one group mean different).

Assumptions: normality, homoscedasticity, independence. One-way ANOVA (single factor) vs Two-way (two factors).

Worked numerical — t-test

Sample of 10 cricket bats: mean weight = 1.18 kg, s = 0.04 kg. Test if population mean = 1.2 kg (α = 0.05, two-tailed).

H_0: μ = 1.2; H_1: μ ≠ 1.2. t = (1.18 − 1.20)/(0.04/√10) = −0.02/0.01265 = **−1.58**. Critical t at df = 9, α/2 = 0.025: ±2.262. |−1.58| < 2.262 → **Fail to reject H_0**: mean = 1.2 kg statistically.

Worked numerical — χ^2 test of independence

2×2 table of gender × preference. Compute O, E (= row tot × col tot / grand tot), then $\chi^2 = \Sigma(O-E)^2/E$. Compare with $\chi^2_{critical}$ at df = (r−1)(c−1). Yates' correction $(|O-E|-0.5)^2$ for 2×2 with small expected freq.

SECTION 3 — CLRM Assumptions & Multiple Regression

3.1 Classical Linear Regression Model (CLRM)

Model: $Y_i = \beta_0 + \beta_1 X_i + u_i$, where Y = dependent, X = independent (regressor), β_0 = intercept, β_1 = slope, u = stochastic / disturbance term capturing omitted variables and randomness. Linear regression = linear in **parameters** (not necessarily in variables).

Seven Classical (Gauss-Markov) Assumptions

(1) Linearity: model linear in β's.

(2) X is non-stochastic / fixed in repeated sampling; or uncorrelated with u.

(3) Zero mean of error: $E(u_i) = 0$ — average shock has no systematic effect.

(4) Homoscedasticity: $Var(u_i) = \sigma^2$ (constant for all i). Violation → heteroscedasticity (OLS unbiased but inefficient).

(5) No autocorrelation: $Cov(u_i, u_j) = 0$ for $i \neq j$. Violation common in time series.

(6) Normality of error: $u_i \sim N(0, \sigma^2)$ — needed for inference (t, F tests).

(7) No perfect multicollinearity (relevant for multiple regression): regressors not perfectly linearly related.

BLUE — Gauss-Markov Theorem

Under assumptions (1)–(5), OLS estimators are **BLUE**: **B**est **L**inear **U**nbiased **E**stimators. • **Linear** in Y; • **Unbiased**: $E(\beta) = \beta$; • **Best**: minimum variance among linear unbiased estimators.

Normality (6) needed only for hypothesis testing / interval estimation, not for BLUE property.

3.2 Multiple Regression Model / बहु प्रतगिमन

Model: $Y_i = \beta_0 + \beta_1 X_{1_i} + \beta_2 X_{2_i} + \ldots + \beta_k X_{k,i} + u_i$. In matrix form: $\mathbf{Y = X\beta + u}$, where Y (n×1), X (n×k+1), β ((k+1)×1), u (n×1).

OLS estimation — matrix form

OLS minimises $\mathbf{u'u} = (Y - X\beta)'(Y - X\beta)$. First-order condition gives:

$\boldsymbol{\beta = (X'X)^{-1} X'Y}$ — the celebrated OLS estimator.

Variance: $Var(\beta) = \sigma^2 (X'X)^{-1}$. Each β_j measures change in Y for a unit change in X_j, *holding all other X's constant* (ceteris paribus partial effect).

Goodness of fit

$\mathbf{R^2 = ESS / TSS = 1 - RSS/TSS} \in [0, 1]$. % of variation in Y explained by regressors.

Adjusted R^2 $= 1 - (1-R^2)(n-1)/(n-k-1)$. Penalises addition of irrelevant regressors. Always prefer adj-R^2 for multiple regression model comparison.

Overall F-test: $H_0: \beta_1 = \beta_2 = \ldots = \beta_k = 0$. $F = (R^2/k) / [(1-R^2)/(n-k-1)]$.

3.3 Common Violations & Diagnostics

Problem	Detection	Consequence & Remedy
Multicollinearity	High R^2 but few t-significant; VIF > 10; high pairwise r	Std errors blow up; signs reverse. Drop variable, transform, ridge regression, PCA.
Heteroscedasticity	Residual plot fan-shape; Breusch-Pagan, White, Goldfeld-Quandt tests	OLS unbiased but inefficient; t/F invalid. Use WLS / GLS / robust SE (White).
Autocorrelation	Durbin-Watson statistic (≈ 2 ok; < 1.5 positive AC); Breusch-Godfrey	Underestimates SE; spurious significance. Use Cochrane-Orcutt / Newey-West SE.
Non-normality of u	Jarque-Bera test (skewness + kurtosis combined)	CLT helps in large samples; transform Y; add omitted regressors.

Problem	Detection	Consequence & Remedy
Specification error	Ramsey RESET test; theory check	Omitted variable bias; include relevant X, correct functional form.

EXAM TIP — Common BLUE / CLRM question

"What is the difference between linear and non-linear regression?" — **Linear regression = linear in parameters** ($Y = \beta_0 + \beta_1 X^2 + u$ is still linear). Non-linear models like $Y = \beta_0 X^{\beta_1}$ require non-linear estimation (or log-transform). Always state the 7 CLRM assumptions when asked about OLS properties.

SECTION 4 — Calculus & Matrix in Consumer, Producer & Market Theory

4.1 Calculus in Consumer Behaviour / उपभोक्ता व्यवहार

Utility function: U = U(x, y). **Marginal Utility** = partial derivative: $MU_x = \partial U/\partial x$. **Law of Diminishing MU**: $\partial^2 U/\partial x^2 < 0$.

Consumer optimisation — Lagrange method

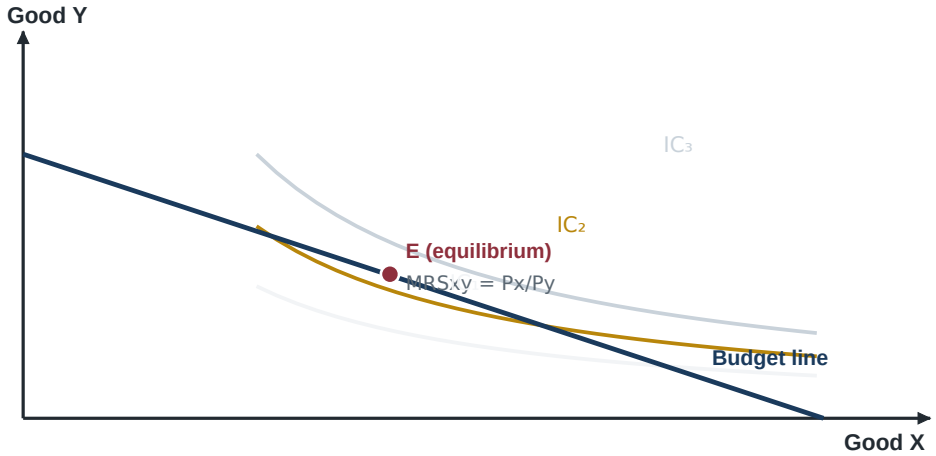
Problem: Maximise U(x, y) subject to $P_x \cdot x + P_y \cdot y = M$.

Lagrangian: $L = U(x, y) - \lambda(P_x \cdot x + P_y \cdot y - M)$.

FOC: $\partial L/\partial x = MU_x - \lambda P_x = 0$; $\partial L/\partial y = MU_y - \lambda P_y = 0$.

Dividing: $\mathbf{MU_x/MU_y = P_x/P_y \rightarrow MRS_{xy} = P_x/P_y}$ (slope of IC = slope of budget line at equilibrium).

Equimarginal Principle: $MU_x/P_x = MU_y/P_y = \lambda$ (= marginal utility of income).



Calculus: max U(x,y) s.t. $P_x x + P_y y = M \rightarrow$ Lagrange $\rightarrow MU_x/MU_y = P_x/P_y$

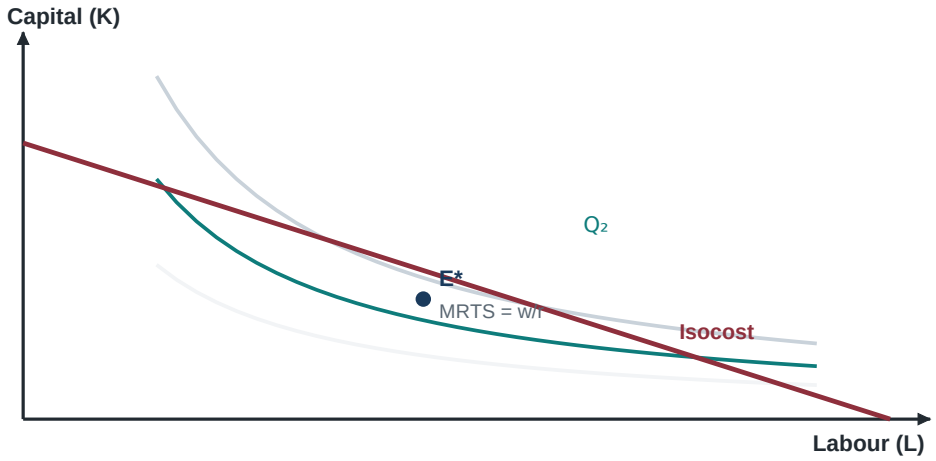
Fig 13: Consumer equilibrium — IC tangent to budget line

Numerical — Optimal consumption

$U = x^{0.5}y^{0.5}$ (Cobb-Douglas), $P_x = 4$, $P_y = 5$, $M = 100$. For C-D: $x^* = (\alpha M)/((\alpha+\beta)P_x)$, $y^* = (\beta M)/((\alpha+\beta)P_y)$. $\alpha = \beta = 0.5 \rightarrow$ **$x^* = 100/8 = 12.5$; $y^* = 100/10 = 10$**. Check: $4\times12.5 + 5\times10 = 50+50 = 100$ ✓

4.2 Calculus in Production

Production function: Q = f(L, K). **Marginal Products**: $MP_L = \partial Q/\partial L$, $MP_K = \partial Q/\partial K$. **$MRTS_{LK}$** = MP_L/MP_K = slope of isoquant.



min Cost = wL + rK s.t. $Q(L,K) = Q_0 \rightarrow MP_l/w = MP_k/r$ (equimarginal)

Fig 14: Producer equilibrium — isoquant tangent to isocost

Producer optimisation

Min cost: C = wL + rK s.t. Q(L, K) = Q_0. Lagrange → **$MP_L/w = MP_K/r$** (least-cost combination).

Returns to scale (Cobb-Douglas $Q = AL^{\alpha}K^{\beta}$): $\alpha + \beta > 1$ IRS; $\alpha + \beta = 1$ CRS; $\alpha + \beta < 1$ DRS.

Euler's product exhaustion theorem: under CRS and perfect competition, $L \cdot MP_L + K \cdot MP_K = Q$ — total product fully distributed to factors at marginal products.

4.3 Calculus in Pricing / Market Structures

Profit maximisation — universal rule

Profit: $\pi(Q) = TR(Q) - TC(Q)$. Setting $d\pi/dQ = 0$ → **MR = MC**. Second order condition: $d^2\pi/dQ^2 < 0$ → MC slope > MR slope.

Perfect competition: P = MR = AR (horizontal demand) → produce where P = MC.

Monopoly: P > MR; using elasticity: $MR = P(1 - 1/|e_p|)$ → $P = MC / (1 - 1/|e_p|)$ (Lerner index).

Price discrimination (3rd degree): $MR_1 = MR_2 = MC$. Higher price in market with less elastic demand.

Matrices in economics

Solving linear systems (market equilibrium for n goods): $AX = B \rightarrow X = A^{-1}B$. Cramer's rule: $x_i = |A_i| / |A|$.

Used in: Input-Output (Leontief), simultaneous equation models, regression OLS $\beta = (X'X)^{-1}X'Y$, game theory payoff matrix, transportation problem.

Hessian matrix of second partial derivatives → second-order conditions for max/min (definite/semi-definite check).

Integral calculus uses

- **From MR to TR**: $TR = \int MR(Q)\,dQ + C$. Similarly TC from MC.
- **Consumer surplus** = $\int_0^{Q^*} [P(Q) - P^*]\,dQ$ (area between demand curve and price line).
- **Producer surplus** = $\int_0^{Q^*} [P^* - MC(Q)]\,dQ$.
- **Capital accumulation**: $K(t) = K_0 + \int_0^t I(s)\,ds$ (gross investment integrated over time).

SECTION 5 — Input-Output Analysis & Linear Programming

5.1 Input-Output Analysis / आगत-नर्गित वश्लेषण

Wassily Leontief (1936, Nobel 1973) developed Input-Output (I-O) analysis — a general equilibrium framework showing inter-industry flows: how each sector's output becomes another sector's input. India me Planning Commission and CSO have constructed I-O tables since 1951-52.

	Agri (A)	Industry (I)	Services (S)	Final	Output
Agri (A)	a_{n11}	a_{n12}	a_{n13}	F_i	X_i
Industry (I)	a_{n21}	a_{n22}	a_{n23}	F_i	X_i
Services (S)	a_{n31}	a_{n32}	a_{n33}	F_i	X_i

$X = AX + F \rightarrow X = (I - A)^{-1} F$ (Leontief inverse — total req. per unit final demand)

aij = inputs from industry i needed per unit of industry j's output (technical coefficient)

Fig 15: Input-Output transactions table — Leontief framework

Mathematical structure

Technical coefficient: $a_{ij} = x_{ij} / X_j$ (amount of i's output needed per unit of j's output). Assumed constant (fixed-proportions, no substitution).

System: $X_i = \Sigma_j a_{ij}X_j + F_i$ (output i = intermediate use + final demand). Matrix form: **X = AX + F**.

Solution: $X = (I - A)^{-1} F$. The matrix $\mathbf{(I - A)^{-1}}$ is the **Leontief inverse** — each element shows total (direct + indirect) requirement of sector i to produce one unit of final demand in sector j.

Hawkins-Simon conditions for non-negative solution: all principal minors of (I−A) positive.

Assumptions and limitations

Assumptions: (a) constant a_{ij} (no technological change); (b) constant returns to scale; (c) no joint production; (d) homogeneous sectors; (e) prices fixed.

Uses: national planning (sectoral targets), impact analysis (multiplier — output, employment, income), inflation analysis, environmental accounting (CO_2 I-O).

Limitations: static; aggregation problem; data-hungry; assumes no substitution among inputs.

5.2 Linear Programming / रैखकि प्रोग्रामगि

Origins: G. B. Dantzig (1947) developed the **Simplex method**; **L. V. Kantorovich** contributed parallel ideas (Nobel 1975 with Koopmans). LP optimises a **linear** objective function subject to **linear** equality / inequality constraints with non-negative variables.

General LP formulation

Maximise (or Min): $Z = c_1x_1 + c_2x_2 + \ldots + c_nx_n$ (objective function)

Subject to: $a_{i1}x_1 + a_{i2}x_2 + \ldots \leq$ (or $\geq$, =) b_i (constraints, i = 1, 2, …, m)

Non-negativity: $x_j \geq 0$ for all j. In matrix form: Max Z = c'x s.t. $Ax \leq b, x \geq 0$.

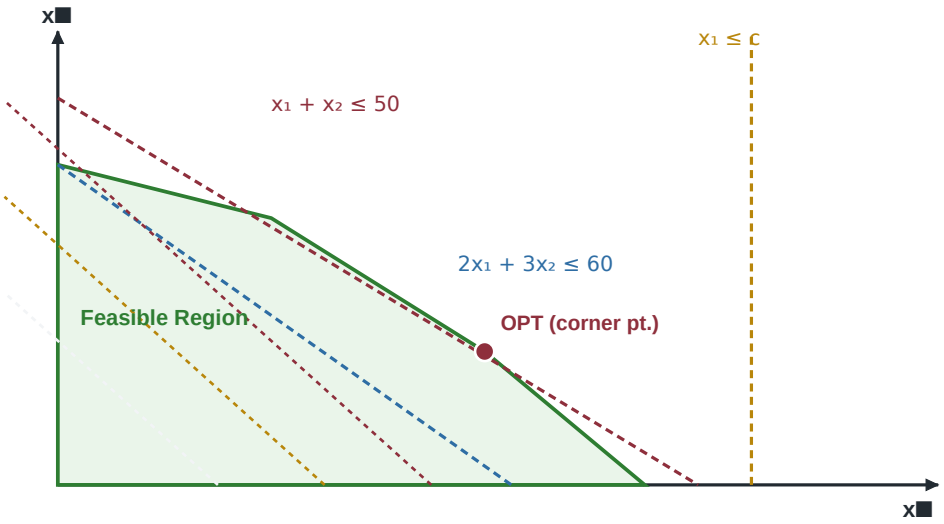


Fig 16: LP graphical solution — feasible region & optimal vertex

Key concepts

Feasible region = intersection of all constraint half-planes (convex polygon).

Theorem: If an optimum exists, it occurs at a **corner point (vertex / extreme point)** of the feasible region — this is the foundation of the Simplex algorithm.

Solution methods: (a) **Graphical** (2-variable); (b) **Simplex** (n-variable, iterative pivot on the simplex tableau); (c) Interior-point (Karmarkar 1984).

Special cases: unique optimum, alternate optima, unbounded, infeasible, degeneracy.

Duality: every LP (primal) has a **dual**: Min b'y s.t. A'y ≥ c, y ≥ 0. Strong duality: optimal values equal. Shadow prices = dual variables.

Worked numerical — LP graphical

A firm produces two products A (profit ₹40/unit) and B (₹30/unit). Each A needs 2 hrs labour, 1 hr machine; each B needs 1 hr labour, 1 hr machine. Labour available: 40 hrs; machine: 30 hrs. Find optimal product mix.

LP: Max Z = 40A + 30B s.t. 2A + B ≤ 40, A + B ≤ 30, A, B ≥ 0.

Corners: (0,0), (20,0), (10,20), (0,30). Z values: 0, 800, 1000, 900. **Optimum at (10, 20): Z = ₹1000**.

Applications of LP in economics

- **Diet problem** (Stigler 1945): cheapest food basket meeting nutritional constraints.
- **Transportation problem**: shipping goods from sources to destinations at minimum cost.
- **Assignment problem**: assigning workers to jobs (Hungarian method).
- **Production planning**, crop allocation, portfolio optimisation, network flows.

QUICK REVISION ANCHORS — एक नज़र में

Concept	Key formula / fact
Empirical relation (skewed dist.)	**Mode = 3 Median − 2 Mean**; AM ≥ GM ≥ HM
Karl Pearson Skewness	Sk = (Mean − Mode) / σ; range typically −3 to +3
Bowley's Skewness	$(Q_3+Q_1-2M) / (Q_3-Q_1)$; range −1 to +1
Kurtosis (β_2)	> 3 lepto; = 3 meso (normal); < 3 platy
Coefficient of Variation	CV = $(\sigma/\bar{X}) \times 100$; relative dispersion
Karl Pearson r	$r = \Sigma xy/\sqrt{(\Sigma x^2 \cdot \Sigma y^2)}$; range [−1, +1]; $r^2 = b_{yx} \cdot b_{xy}$
Spearman ρ	$1 - 6\Sigma d^2/[n(n^2-1)]$
Probable Error	$PE = 0.6745(1-r^2)/\sqrt{n}$; r > 6·PE significant
Regression coefficients	$b_{yx} = r \cdot \sigma_y/\sigma_x$; $b_{xy} = r \cdot \sigma_x/\sigma_y$
Fisher's Ideal Index	$F = \sqrt{(L \times P)}$; passes time + factor reversal
Time series	Y = T + C + S + I (additive) or T × C × S × I (multiplicative)
Binomial	X~B(n,p); E(X)=np, Var=npq
Poisson	$P(X=r) = e^{-\lambda}\lambda^r/r!$; E=Var=λ
Normal — empirical rule	68.27% within μ±σ; 95.45% within μ±2σ; 99.73% within μ±3σ
Z-score	$Z = (X - \mu) / \sigma$; Z ~ N(0, 1)
CLT	$\bar{X} \sim N(\mu, \sigma^2/n)$ for large n irrespective of parent distribution
Type I / II error	$\alpha = P(\text{Reject } H_0 \mid \text{true})$; $\beta = P(\text{Accept } H_0 \mid \text{false})$; Power = 1 − β
t-test df	df = n − 1 (one sample); $n_1 + n_2 - 2$ (two sample)
χ^2 df (contingency)	df = (r − 1)(c − 1)
F-statistic ANOVA	F = MSB/MSW; df = (k−1, n−k)
OLS (BLUE)	$\beta = (X'X)^{-1}X'Y$; Best Linear Unbiased Estimator under Gauss-Markov
R^2 vs adj R^2	R^2 = ESS/TSS; adj R^2 penalises extra regressors
Durbin-Watson	≈ 2 → no AC; < 1.5 positive AC; > 2.5 negative AC
Multicollinearity	VIF > 10 indicates problem
Consumer eq.	$MU_x/MU_y = P_x/P_y$ (MRS = price ratio)
Producer eq.	$MP_L/w = MP_K/r$ (least cost)
Profit max	$d\pi/dQ = 0 \rightarrow MR = MC$; $d^2\pi/dQ^2 < 0$
Lerner Index	$(P-MC)/P = 1/\lvert e_p \rvert$
Input-Output	$X = (I - A)^{-1} F$ (Leontief inverse)
LP	Max Z = c'x s.t. Ax ≤ b, x ≥ 0; optimum at vertex (Simplex)
LP Duality	Primal Max ↔ Dual Min; shadow prices = dual variables

Last-minute exam strategy

(1) Likhne ka order: Definition → Formula → Diagram → Numerical → Interpretation. Statistics me numerical worth at least 50% of the answer's weight.

(2) Hamesha state the assumptions before discussing OLS/ANOVA/t-test — examiners weight assumptions heavily.

(3) Sketch a small diagram for any distribution / test / consumer-producer eq. (5 min, +5 marks).

(4) Connect statistics to economics: e.g., when discussing CLRM, mention demand estimation; for I-O mention India's sectoral planning.

(5) Keep one tested numerical for each of: mean/SD/CV, r and regression, χ^2 2×2, ANOVA F, normal Z-prob, LP graphical.